

Tony Smoragiewicz

 24/03/1994 Toledo, OH, USA
 USA citizen
 tonysmoragiewicz@gmail.com
 tony-smoragiewicz
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EDUCATION

- 2025 - 2026 **Technical University of Munich, München, DE**
PhD candidate at the Munich Institute of Robotics and Machine Intelligence
Researching applications for controls and computer vision in exoskeletons under Prof. Lorenzo Masia
- 2021 - 2023 **Northeastern University, Boston, MA, USA**
Master of Science in Robotics with concentration in Computer Science
- 2012 - 2017 **University of Michigan, Ann Arbor, MI, USA**
Bachelor of Science in Engineering - Aerospace Engineering

EXPERIENCE

- April. 2026 - **Robotics Software Engineer, Agile Robots SE**
Present
 - Created and maintained remote over-the-air updates for dozens of robots around the world
 - Led a wireless connectivity conversion that cut latency in half for robot teleoperation
 - Deployed my first software product within two months at the company
- April. 2025 - **PhD Candidate, TU Munich**
March 2026
 - Project lead for the software stack of our Python-based exosuit on the NVIDIA Jetson
 - Created and managed CI/CD pipelines and a GitHub Organization site for the entire lab
 - Taught software best practices for Git and Python to researchers to improve code reliability
- Mar. 2024 - **Aerospace Algorithms Engineer, MORSE Corp.**
Mar. 2025
 - Wrote safety-critical localization software for post-processing time series data of new NATO aircraft
 - Reduced state estimation during GPS-denied flight by 100's of meters during flight
 - Mentored interns on software architecture, code review, testing and design patterns
 - Implemented computer vision tools for scanned document alignment with point clouds
 - Presented engineering solutions to our customers in the armed forces
- Sep. 2022 - **Graduate Research Assistant, US Army Research Laboratory**
Feb. 2024
 - Wrote nonlinear optimization algorithms for localizing a UAV within a known environment
 - Built software to verify the accuracy of visual SLAM algorithms on an NVIDIA Jetson Nano
 - Created a novel Monte Carlo simulation that improved munition state estimation by 400%
 - Parallelized an existing probabilistic dynamical models which had a 530% reduction in runtime
- Mar. 2018 - **Professional Triathlete & Engineer, United States National Team**
Jun. 2021
 - Designed and created novel racing equipment to improve my competitive advantage
 - Modeled dynamical race simulations from sensor data to improve equipment selection
 - Represented the USA at the Olympic Test event in Tokyo, Japan
 - Earned a silver medal at the 2019 Cape Town World Cup in South Africa
 - Ranked top 40 in the world at the highest point in my athletic career
- Jan. 2017 - **R&D Engineer, Specialized Bicycle Components**
Mar. 2018
 - Wrote embedded algorithms in C++ to track athletes' movement patterns in the field of play
 - Saved the company over \$50k by designing a new thermal data acquisition unit for helmet tests
 - Used laser scanner data for custom-fit products to prototype manufacturing options

- Jun. 2016 - **Intern, Specialized Bicycle Components**
- Dec. 2016
- Created a UI and database to easily compare thousands of previous wind tunnel tests
 - Developed model fitting software to measure rotational drag for wheels without strain gauges
 - Wrote C++ firmware for brushed DC motors, reed switches, and thermistors
 - Wrote MATLAB software to automate wind tunnel testing input by 70% during wind tunnel testing
- Jan. 2016 - **Intern, Alphamantis Technologies**
- Apr. 2016
- Wrote C++ software that automated calibration for the Aerosensor (outdoor aerodynamic testing)
 - Diagnosed hardware deficiencies that limited the environmental operating range of the Aerosensor

ACADEMIC PROJECTS

- 2023 **Obstacle Avoidance with Model Predictive Control, Independent**
Created a longitudinal flight dynamics simulator with a labmate for his thesis. We used convex optimization in CVXPY to find optimal rudder and throttle commands for obstacle avoidance. The simulator replicated a noisy LIDAR sensor for obstacle detection and copied the OpenAI Gym environment.
- 2022 **Robotic Pick-and-Place, NEU Computer Science Dept. - Prof. Lawson L.S. Wong**
Wrote unsupervised machine learning algorithms to detect the 6D pose of objects from a structured light camera to improve object manipulation for a Trossen Interbotix arm for a semester-long project.
- 2021 **Vertical Take-Off and Landing Drone, Aerospace NU**
Wrote an open source program to design a VTOL drone. The program aids in the preliminary design phase and is limited to a 4kg payload over 100km. The model handles physical constraints and FAA regulations.
- 2015 **University of Michigan Wind Tunnel, Undergraduate Researcher**
Investigated solutions to remove unwanted turbulence from the 5ft-by-7ft subsonic wind tunnel. My team mapped the test section volume with a hot wire anemometer and we won an award for our paper.
- 2014 **Remote Controlled Hovercraft, UM Aerospace Dept. - Prof. Peter D. Washabaugh**
Designed and constructed a remote controlled hovercraft for a semester long project. Our team was commended for original passive energy capturing designs which hadn't been seen from previous teams.

AWARDS

- 2016 **Landes Contest for Technical Communication, Second Place**
My team's final report for our senior project won a top spot at a university-led writing competition. We presented our findings for how the University could improve the aerodynamic qualities of campus' wind tunnel. Our methods were cost effective and addressed hazardous materials used in original construction.
- 2014 & 2015 **Michigan Sportsmanship Award, Winner**
Selected by the athletic department for exemplary sportsmanship and ethical behavior.
- 2014 **Michigan Leadership Academy, Invited Member**
Selected for a personal development course to help student-athletes become effective leaders on campus. During the course we learned leadership skills such as about conflict resolution, methods to reduce hazing, and ways to address issues during power imbalances.
- 2014 **Intro to Computers and Programming, Winner**
Awarded the best algorithm for a path finding competition designed to help a search and rescue robot.

TECHNICAL SKILLS

Languages	Python, C++, PyTorch, MATLAB, ROS, Bash
Development	Git, Conda, Poetry, Jupyter Notebooks, Taskfiles
Sensors	Stereo Camera, LIDAR, RTK GPS, IMU/AHRS/MARG, Structured Light Camera
Fabrication	Laser Cutter, 3D Printer, CNC Machine, Lathe

LANGUAGES

English, mother tongue
German, B1 and actively learning

TEACHING & COMMUNITY ENGAGEMENT

- Jun. 2022 - **Polygence, Robotics Mentor**
current Guided numerous international high school students through their own research projects. These range from using PyTorch to train robotic cars, creating a computer vision-powered bicycle light that alerts the rider of approaching threats, to creating regression models to improve AprilTag detection.
- Feb. 2023 - **Boston Youth Farming Project, Classroom Educator**
Dec. 2023 Worked with local elementary schools to teach children about nutrition, the environment, and plant biology with a big focus on student empowerment. Provided hands-on experience in urban farming.
- Sep. 2023 - **South Boston Catholic Academy, 6th Grade Science and Robotics Club**
Feb. 2024 Created novel teaching material to show younger students how complex electrical components can be understood without advanced mathematics. Taught students how to use CAD and a 3D printer.